Database Matchmaking

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Accessibility Notice

This document is also available in HTML format at:

Link: `https://aholdengouveia.name/AdvData/labs/wk9matchmaking.html`

The HTML version provides enhanced accessibility features including keyboard navigation, screen reader support, responsive design, dark mode support, and high contrast options.

Request

Request: DATA-1487
Requested by: Finance director
Data: Your database from the Beyond SQL lab

Background

The company is growing and may start collecting new kinds of data. Before spending any money, the finance director wants a recommendation: should the company keep its relational database, or move some or all of its data to a NoSQL database? The director controls the budget but isn't technical, so your recommendation has to be clear about costs, risks, and trade-offs.

Learning Outcomes

By the end of this lab, you will be able to:

- Evaluate relational and NoSQL database types for a data set using business needs
- Identify new data a company could collect, and how it affects the choice of database
- Compare database vendors, including free and paid options and support
- Recommend a database, and justify it to a non-technical decision maker

Lab Instructions

Use your own data set and the company from your data card or the Database Documentation lab. Books are my example data only.

Part 1: Your Company's Needs

Answer these questions for your company. Use the size you gave your company in the Database Documentation lab.

1. How much data does the company have, and how fast is it growing?
2. How many people add data, and how many only look at it?
3. What is the budget for the database, including staff time and support?
4. How much can the company afford for the database to be down?
5. What tools or systems does the database need to work with?
6. What skills does the team already have?
7. Does the company have to follow any rules about its data, such as privacy laws?

Part 2: New Data

List at least 5 new kinds of data your company could collect in the future. For each one, explain how it would be collected, and whether it would fit neatly into tables or not.

Part 3: Evaluate the Options

For each of these 5 options, give at least 3 pros and 3 cons for your company, based on your answers to Parts 1 and 2:

1. Your current relational database
2. Key value store
3. Document store
4. Graph database
5. One other NoSQL type of your choice, such as wide-column, object, tuple store, or triplestore

Part 4: Compare Vendors

For your top 2 options from Part 3, pick a vendor for each, such as PostgreSQL, MongoDB, Redis, or Neo4j. For each vendor, explain:

1. Whether it's free, paid, or has both a free and a paid version
2. What support is available, and whether it costs extra
3. Why it would suit your company

Cite your sources with links, and check each one with the CRAAP test (linked under Resources). A vendor's own website is fine for facts like prices, but remember it's trying to sell you something.

Part 5: Recommendation Memo

Write a memo of about one page to the finance director, following the sample recommendation memo (linked under Resources). Put your recommendation first, then explain why, the cost, the risks, what happens if the company does nothing, and the next step. Use plain words, and save the technical details for Parts 1 to 4.

Acceptance Criteria

This request is done when:

- All 7 questions about your company's needs are answered
- At least 5 new kinds of data, each with how it would be collected
- All 5 options have at least 3 pros and 3 cons, specific to your company
- 2 vendors are compared, with cited sources that you've checked with the CRAAP test
- Your memo leads with a recommendation, is about one page, and could be understood by someone who isn't technical

Deliverables

One file that includes Parts 1 to 5.

Audience: the finance director, who controls the budget but isn't technical. Parts 1 to 4 can be technical, since the data team will check them.

Resources

References, a video, a PowerPoint and some notes are available at my website Link: `https://www.aholdengouveia.name/AdvData/Databasematchmaking.html`

NoSQL types, on the Beyond SQL topic page Link: `https://www.aholdengouveia.name/AdvData/BeyondSQL.html`

Sample recommendation memo, showing the format to follow Link: `https://www.aholdengouveia.name/AdvData/labs/SampleRecommendationMemo.txt`

The CRAAP test, for checking your sources Link: `https://web.archive.org/web/20250814000638/https://guides.lib.uchicago.edu/c.php?g=1241077&p=9082343`

My guide to evaluating sources Link: `https://www.aholdengouveia.name/Howtos/Sourceeval/sourceEvaluation.html`